

BATCH No:001

DESIGN AND DEVELOPMENT OF SMART CHAIR

*Minor Project-II report submitted
in partial fulfillment of the requirement for award of the degree of*

**Bachelor of Technology
in
Information Technology**

By

Sabthika Preethy	(23UEIT0050)	(VTU24624)
M.Deepika	(23UEIT0035)	(VTU26653)
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*Under the guidance of
Dr. VISUMATHI.J, M.E,Ph.D.,
PROFESSOR*



**DEPARTMENT OF INFORMATION TECHNOLOGY
SCHOOL OF COMPUTING**

**VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF
SCIENCE AND TECHNOLOGY**

(Deemed to be University Estd u/s 3 of UGC Act, 1956)

**Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA**

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CERTIFICATE

It is certified that the work contained in the project report titled "DESIGN AND DEVELOPMENT OF SMART CHAIR" by "SABTHIKA PREETHY (23UEIT0050), M.DEEPIKA (23UEIT0035), V.MAMATHA (23UEIT0059)" has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We hereby declare that the project report entitled “Design and Development of Smart Chair” is a result of our own work carried out under the guidance of our project supervisor as part of our academic project work. We further declare that this written submission represents our own work and original ideas, and wherever the ideas or words of others have been included, they have been properly cited and acknowledged. We have followed all principles of academic honesty and integrity, and we understand that any violation of these principles will invite disciplinary action by the Institute and may also result in penal consequences from the original sources that have not been properly credited..

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APPROVAL SHEET

This project report entitled DESIGN AND DEVELOPMENT OF SMART CHAIR by SABTHIKA PREETHY (23UEIT0050), M.DEEPIKA (23UEIT0015), V.MAMATHA (23UEIT0059) is approved for the degree of B.Tech in Information Technology.

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ABSTRACT

Prolonged sitting and improper posture are major causes of musculoskeletal disorders, back pain, fatigue, and reduced productivity in modern lifestyles. This project, “Design and Development of a Smart Chair,” aims to develop an intelligent and automated seating system that promotes healthy sitting habits through continuous monitoring and real-time feedback. The system is structured into three main modules: Data Acquisition Module, Data Processing and Feature Extraction Module, and Control Logic Alert Module. In the data acquisition stage, a load cell integrated with the HX711 amplifier is used to measure the user’s weight and accurately detect the presence of a person on the chair, while the MPU6050 sensor captures posture-related data such as acceleration and orientation, which helps in identifying body tilt and movement. The collected sensor data is transmitted to the ESP32 microcontroller, where it is processed and converted into meaningful parameters such as tilt angles and sitting duration. In the processing stage, the system applies basic filtering techniques to obtain stable readings and extracts important features like posture angle and time of sitting. The control logic module implements threshold-based detection, posture analysis, and timer-based monitoring algorithms to evaluate user behavior. Based on predefined threshold values, the system determines whether the posture is correct or incorrect and checks if the sitting duration exceeds a safe limit. If any abnormal condition such as slouching, leaning, or prolonged sitting is detected, the system immediately generates an alert using a buzzer to notify the user. The system operates continuously in real time, ensuring quick response and accurate monitoring. This smart chair is designed to be cost-effective, simple to implement, and user-friendly, making it suitable for applications in offices, educational institutions, and home environments. The project demonstrates the practical application of embedded systems and sensor-based monitoring in improving ergonomics, enhancing user comfort, preventing long-term health issues, and encouraging better sitting habits in daily life.

Keywords:

Smart Chair, Posture Monitoring, Pressure Sensors, Embedded Systems, Microcontroller, Ergonomics, Health Monitoring, Sitting Behavior Analysis, Real-Time Feedback, Buzzer Alert System

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LIST OF ACRONYMS AND ABBREVIATIONS

S.no	abbr	Abbreviations
1.	ESP32	Espressif 32-bit Microcontroller
2.	MPU6050	Motion Processing Unit Sensor
3.	HX711	Load Cell Amplifier Module
4.	DC	Direct Current
5.	I2C	Inter-Integrated Circuit
6.	Kg	Kilogram
7.	IoT	Internet of Things
8.	USB	Universal Serial Bus
9.	Wi-Fi	Wireless Fidelity

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Chapter 1

INTRODUCTION

1.1 Introduction

In the modern digital era, individuals spend a significant amount of time sitting while working, studying, or using electronic devices. This prolonged sitting, often combined with improper posture, has led to an increase in health issues such as back pain, neck strain, and musculoskeletal disorders. Poor sitting habits not only affect physical health but also reduce concentration, productivity, and overall well-being. Traditional chairs do not provide any mechanism to monitor or correct posture, making users unaware of their incorrect sitting positions until discomfort arises. Hence, there is a growing need for a smart solution that can help maintain proper posture and promote healthy sitting habits. The Design and Development of a Smart Chair aims to address this problem by creating an intelligent seating system that continuously monitors the user's posture and sitting duration. The system uses sensors such as pressure sensors and posture detection modules to track body alignment and weight distribution. A microcontroller processes the sensor data and identifies incorrect posture or prolonged sitting. Whenever such conditions are detected, the system provides immediate feedback through visual indicators like audible alerts using a buzzer, enabling users to correct their posture in real time. This project focuses on developing a cost-effective, user-friendly, and efficient system without relying on complex external applications. The smart chair can be used in offices, educational institutions, and home environments to improve ergonomics and user comfort. By encouraging proper posture and regular movement, the system helps reduce health risks associated with sedentary lifestyles. Overall, this project demonstrates the effective use of embedded systems in enhancing daily life and promoting better health and productivity.

1.2 Background

Prolonged sitting and poor posture have become increasingly common in today's technology-driven lifestyle, leading to various health issues such as back pain, neck strain, and musculoskeletal disorders. Many individuals remain seated for extended periods without being aware of their posture, which gradually affects their physical well-being and productivity. Conventional chairs do not provide any mechanism to monitor or guide users, making it difficult to maintain proper ergonomic practices throughout the day. There is a growing need for an intelligent system that can assist users in improving their sitting habits.

The concept of a smart chair has emerged as an effective solution to promote better posture and reduce health risks associated with sedentary behavior. By integrating sensing technologies and embedded systems, the chair can continuously monitor the user's sitting position and duration. The collected data can be analyzed in real time to detect improper posture or prolonged sitting, enabling timely corrective actions. This approach helps in creating awareness among users and encourages them to maintain a healthier sitting routine.

The development of a smart chair focuses on simplicity, efficiency, and accessibility, making it suitable for everyday use in various environments such as offices, educational institutions, and homes. By providing immediate feedback through simple alert mechanisms, the system supports users in improving posture and taking necessary breaks. This project highlights the importance of ergonomics and demonstrates how technology can be effectively used to enhance comfort, productivity, and overall well-being.

The proposed system emphasizes a cost-effective and user-friendly design that does not rely on complex external interfaces or applications. The use of a microcontroller along with sensors and alert components ensures reliable performance while keeping the system simple to implement and maintain. This makes the smart chair a practical and scalable solution that can be widely adopted to promote healthier lifestyles and reduce the long-term impact of sedentary habits.

1.3 Objective

The primary objective of the Design and Development of a Smart Chair is to create an intelligent seating system that helps users maintain proper posture and reduce the health risks associated with prolonged sitting. The project aims to monitor the user's sitting posture and duration continuously using sensors such as pressure sensors and posture detection modules. By processing this data through a microcontroller, the system can accurately identify incorrect posture and extended periods of inactivity.

The objective is to provide real-time feedback through simple alert mechanisms such as LEDs and buzzer notifications, enabling users to correct their posture immediately and adopt healthier sitting habits. The project also aims to integrate IoT technology using an ESP32 microcontroller to enable wireless communication and real-time data monitoring. This allows posture data to be transmitted to a mobile device for analysis and tracking. The system is intended to be cost-effective, energy-efficient, and easy to implement using readily available components.

The objective includes making the system scalable for future enhancements such as cloud-based monitoring, AI-based posture classification, and advanced health analytics. The project aims to provide an intelligent, automated, and user-friendly solution for improving posture and enhancing overall health and productivity.

The project focuses on developing a cost-effective, user-friendly, and standalone system that does not rely on complex external applications. The smart chair is designed to be easily deployable in environments such as offices, educational institutions, and homes, making it accessible to a wide range of users.

The project also demonstrates the practical application of embedded systems in solving real-world health-related problems. By improving ergonomics and promoting awareness about proper posture, the smart chair contributes to enhancing user comfort, productivity, and overall well-being.

The main objective of this project is to design and develop a Smart Chair system that promotes correct sitting posture and helps prevent health issues caused by prolonged sitting. The system continuously monitors the user's posture in real time using embedded sensors integrated within the chair. It focuses on detecting incorrect sitting positions and providing instant feedback to the user through alert mechanisms such as a buzzer, encouraging immediate correction of posture.

1.4 Problem Statement

In today's digital and technology-driven environment, individuals spend a significant amount of time sitting while working, studying, or engaging with electronic devices. This prolonged sitting, often accompanied by improper posture, has become a major cause of health issues such as back pain, neck strain, spinal misalignment, and other musculoskeletal disorders. Many people are unaware of their incorrect sitting habits, as traditional chairs do not provide any form of monitoring or feedback. Over time, this lack of awareness leads to chronic health problems, reduced productivity, and decreased overall well-being.

Existing solutions to address posture-related issues are often expensive, complex, or dependent on external devices and applications, making them less accessible for everyday use. There is a lack of simple and effective systems that can continuously monitor sitting posture and provide immediate feedback to the user. Without timely alerts or guidance, users tend to remain in unhealthy positions for long durations. There is a need for a cost-effective, user-friendly, and standalone system that can detect improper posture and prolonged sitting in real time and assist users in maintaining correct ergonomic practices, reducing health risks and improving comfort and productivity.

A major issue is the absence of a simple, cost-effective, and user-friendly system that can be easily integrated into a chair for continuous monitoring. Without timely alerts or corrective guidance, users are likely to maintain poor posture for long durations, increasing the risk of long-term health complications. There is also a lack of data tracking mechanisms that can help users analyze their sitting behavior over time and improve their habits.

There is a strong need to design and develop an intelligent, low-cost, and standalone Smart Chair system that can continuously monitor sitting posture in real time, detect improper positions, and provide immediate feedback to the user. Such a system can enhance awareness, encourage healthy sitting habits, and help in reducing health risks while improving comfort, productivity, and overall quality of life.

Existing posture correction solutions are often limited in functionality, expensive, or require external wearable devices and mobile applications, making them less practical for continuous everyday use.

Chapter 2

LITERATURE REVIEW

A.K. Bhoi et al., [1]: A.K. Bhoi et al. explored IoT-enabled smart healthcare monitoring systems and their applications, challenges, and future trends in the *Journal of Healthcare Engineering*. Their study explains how IoT devices, embedded systems, and wireless sensor networks are transforming healthcare by enabling continuous monitoring and real-time data analysis. The research highlights the integration of cloud computing and communication technologies for efficient health monitoring. It also discusses challenges such as data security, privacy, interoperability, and energy consumption in IoT-based healthcare systems. This work provides a strong foundation for developing smart healthcare devices such as posture monitoring systems and ergonomic smart chair solutions by combining sensing, communication, and intelligent analysis.

M.S. Reddy et al., [2]: M.S. Reddy et al. developed a smart posture monitoring system using IoT and embedded sensors, published in the *International Journal of Intelligent Systems and Applications*. Their research focuses on using pressure sensors and posture detection modules to continuously monitor user sitting posture and provide real-time feedback through alerts and notifications. The system helps users maintain correct posture and reduce health risks caused by prolonged sitting. Their study also emphasizes the importance of user-friendly interfaces and remote monitoring through connected applications. This literature is highly relevant because it demonstrates practical implementation of posture monitoring using embedded technologies and real-time alert systems.

R. Sharma et al., [3]: R. Sharma et al. proposed a real-time sitting posture detection system using machine learning techniques in *Procedia Computer Science*. Their study applies machine learning models to classify sitting postures based on real-time sensor data, improving the accuracy of posture recognition and helping prevent musculoskeletal disorders. The research compares different classification techniques

and concludes that intelligent algorithms can significantly reduce detection errors. It also explains how continuous monitoring and automated classification improve user awareness and corrective action. This literature supports the use of intelligent algorithms such as the Decision Tree and Random Forest in smart chair systems for accurate posture monitoring.

H. Patel et al., [4]: H. Patel et al. designed and implemented a smart chair for posture correction and health monitoring, published in the *International Journal of Advanced Science and Technology*. Their system integrates pressure sensors, buzzers, and embedded controllers to detect poor posture and provide immediate alerts, which improve user awareness and sitting habits effectively. The study demonstrates how embedded systems can be used in ergonomic furniture to improve health and productivity. It also highlights the importance of simple and low-cost designs for large-scale adoption in workplaces and educational institutions. This research strongly supports the importance of real-time feedback mechanisms in smart chair applications.

J. Kim et al., [5]: J. Kim et al. introduced a Random Forest-based posture classification system for ergonomic healthcare applications in the journal *Sensors*. Their research uses multiple decision trees to classify user posture more accurately and reliably, reducing classification errors and improving robustness in posture detection systems. The study explains how ensemble learning techniques outperform traditional single-model classifiers in handling complex sensor data. It also emphasizes the importance of accurate posture classification in preventing long-term health issues and improving user comfort. This work highlights the effectiveness of the Random Forest algorithm in enhancing the accuracy and reliability of smart chair posture monitoring systems.

2.1 Existing System

In the current scenario, most individuals rely on conventional chairs that are designed only for basic seating and comfort, without considering the long-term impact of posture on health. These chairs do not have any built-in mechanism to monitor or guide the user's sitting posture, leading people to remain in incorrect positions for extended periods. Prolonged sitting in improper posture is a major cause of back pain, neck strain, and musculoskeletal disorders, and users often become aware of these issues only after experiencing discomfort or fatigue. Some existing solutions include wearable posture correction devices and advanced ergonomic chairs with smart features; however, these systems are often expensive, complex, and dependent on external devices or applications. Wearable devices can be uncomfortable for continuous use, while advanced chairs may require technical knowledge for setup and maintenance. Many existing systems lack real-time, simple alert mechanisms that can immediately notify users about incorrect posture. They may also consume more power, require frequent calibration, and depend on continuous connectivity, making them less reliable in all situations. These systems are not always customizable to different user needs and body types, which can limit their effectiveness, and they often lack portability, making them difficult to use in different environments. Maintenance costs, hardware complexity, and durability become concerns for long-term usage. In some cases, delayed feedback reduces the effectiveness of posture correction, as users do not receive immediate guidance. Data privacy and security concerns may arise in systems that rely on external connectivity, and the lack of standardization in existing designs makes integration and scalability challenging. There is a need for a cost-effective, user-friendly, and standalone system that can continuously monitor posture and provide instant feedback without relying on complex interfaces or external dependencies, making it suitable for widespread adoption. Existing solutions are often expensive, complex, or not suitable for everyday use in homes, offices, or classrooms. They also lack proper integration of IoT and embedded technologies, which limits their ability to provide smart, automated, and continuous monitoring. Data storage and analysis features are minimal or completely absent, making it difficult to track posture history and improvements over time.

2.2 Related Work

Several research studies have focused on developing smart seating systems to address posture-related health issues caused by prolonged sitting. Early approaches primarily utilized pressure sensors embedded in chairs to monitor weight distribution and detect improper posture. These systems demonstrated that variations in pressure patterns can be effectively used to identify incorrect sitting positions. However, many of these initial designs lacked real-time feedback mechanisms, limiting their ability to actively assist users in correcting their posture. With the advancement of embedded systems, researchers began integrating microcontrollers with various sensors such as force-sensitive resistors and posture detection modules to improve accuracy and responsiveness. These systems enabled continuous monitoring of body alignment and provided immediate feedback through LEDs and buzzer alerts when poor posture was detected. While these solutions improved functionality, some of them were still costly or required complex configurations, making them less accessible for everyday use.

2.3 Research Gap

Despite significant advancements in posture monitoring and smart chair technologies, several limitations still exist in current systems. Many existing solutions focus on high-end designs that incorporate IoT, cloud computing, or mobile applications, which increase system complexity and cost, making them less accessible for general use. Some systems rely on wearable devices that may be uncomfortable for continuous usage and do not always provide consistent monitoring. Many existing approaches lack effective real-time feedback mechanisms; even though they collect and analyze posture data, they often fail to provide immediate alerts that help users correct their posture instantly, reducing their effectiveness. Issues such as high power consumption, frequent calibration requirements, and dependency on continuous connectivity limit the practicality and reliability of these systems in everyday environments. There is a clear need for a cost-effective, user-friendly, and standalone smart chair system that can continuously monitor posture and provide instant feedback without relying on complex external interfaces.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

In the present scenario, most individuals use conventional chairs that are designed only for basic comfort and do not provide any mechanism to monitor or correct sitting posture. As a result, users often sit for long hours in improper positions without being aware of the potential health risks. This leads to various problems such as back pain, neck strain, spinal disorders, and reduced productivity. Although some advanced ergonomic chairs and posture monitoring systems have been introduced, they are not widely adopted due to their high cost and complexity. Many of these systems rely on wearable devices or external applications, which may not be convenient for continuous use. The existing systems suffer from several disadvantages. They are often expensive and not affordable for all users, making them unsuitable for large-scale implementation. Many solutions depend on mobile applications or cloud connectivity, increasing system complexity and requiring continuous internet access. Wearable devices can cause discomfort and may not always provide accurate data if not used properly. Some systems lack real-time feedback, delaying posture correction and reducing effectiveness. Frequent calibration, high power consumption, and maintenance requirements further limit their usability. There is a need for a simple, cost-effective, and standalone solution that can continuously monitor posture and provide immediate feedback, overcoming the limitations of existing systems. Existing solutions are often expensive, complex, or not suitable for everyday use in homes, offices, or classrooms. They also lack proper integration of IoT and embedded technologies, which limits their ability to provide smart, automated, and continuous monitoring. Data storage and analysis features are minimal or completely absent, making it difficult to track posture history and improvements over time.

3.2 Proposed System

The proposed system focuses on the design and development of a Smart Chair that can continuously monitor the user's sitting posture and duration using embedded sensors and a microcontroller. The system integrates pressure sensors and posture detection modules within the chair to analyze body alignment and weight distribution in real time. The collected data is processed by the microcontroller to identify improper posture or prolonged sitting. Whenever such conditions are detected, the system provides immediate feedback to the user through a buzzer notification. This real-time feedback enables users to correct their posture instantly and develop healthier sitting habits without relying on external applications or complex interfaces. The proposed system offers several advantages over existing solutions. It is cost-effective and designed to be affordable for a wide range of users, making it suitable for large-scale implementation in offices, educational institutions, and home environments. The system is designed with flexibility and adaptability in mind, allowing it to be integrated into different types of chairs without major structural modifications. The modular design of the components makes troubleshooting and upgrades easier, enabling future enhancements such as adding wireless communication or mobile-based monitoring if required. The system ensures consistent performance under varying user weights and sitting conditions, maintaining accuracy in posture detection. By encouraging users to take regular breaks and maintain proper posture, it helps in preventing musculoskeletal problems, back pain, and fatigue, thereby improving overall workplace efficiency and comfort. The system promotes awareness about ergonomic practices by providing continuous feedback, which gradually helps users build long-term healthy habits. Its reliability and simplicity make it especially beneficial in environments where continuous supervision is not possible. The absence of complex user interfaces reduces distractions, allowing users to focus on their tasks while still benefiting from automated monitoring. The proposed smart chair enhances ergonomics and user comfort and contributes to improved productivity, reduced health risks, and a better quality of life, making it a practical and impactful solution for modern sedentary lifestyles.

3.3 Feasibility Study

The proposed Smart Chair system is highly feasible from technical, economic, and operational perspectives. From a technical standpoint, the project utilizes readily available components such as pressure sensors, posture detection modules, microcontrollers, and a buzzer, which are easy to integrate and program using standard embedded system techniques. The design does not require complex infrastructure or advanced software, making implementation straightforward, stable, and reliable. The system can be developed using commonly used programming environments, and its modular architecture allows easy debugging, testing, and future enhancements. From an economic perspective, the system is cost-effective as it uses low-cost hardware components, ensuring affordability for students, offices, and home users. The absence of external dependencies such as mobile applications, cloud platforms, or subscription-based services reduces both development and long-term maintenance costs. The components are easily available in the market, which simplifies replacement and reduces downtime. The system's low power consumption contributes to reduced operational expenses, making it suitable for continuous use without increasing electricity costs. From an operational perspective, the system is simple, user-friendly, and requires minimal user interaction. It provides real-time feedback through buzzer alerts, enabling immediate posture correction and timely break reminders. The installation process is straightforward and does not require specialized technical skills, making it convenient for deployment in various environments. The system requires minimal maintenance, as the sensors and microcontroller are durable and do not need frequent calibration. The system demonstrates strong scalability and adaptability, as it can be integrated into different types of chairs such as office chairs, study chairs, and wheelchair designs with minor modifications. Its compact and lightweight design ensures that it does not affect the comfort or usability of the chair. The reliability of the system under different user conditions, such as varying body weights and sitting patterns, further enhances its practical applicability.

3.3.1 Economic Feasibility

The proposed Smart Chair system is economically feasible as it is designed using low-cost and easily available components such as pressure sensors, a microcontroller, and a buzzer. Unlike advanced smart furniture systems that rely on expensive technologies like IoT connectivity, cloud services, or mobile applications, this project avoids additional costs related to software development, internet usage, and data maintenance. The overall development cost is minimal, making it affordable for students, institutions, offices, and home users. The system requires very low maintenance, as the components used are durable and do not need frequent replacement or calibration. The power consumption of the system is also low, which reduces operational expenses over time. Since the design is simple and does not involve complex installation procedures, there are no extra costs for specialized setup or technical support. The scalability of the system allows it to be implemented in multiple chairs at a low incremental cost, making it suitable for large-scale deployment in workplaces and educational environments. The system eliminates the need for recurring subscription-based services or software licensing fees, which are commonly associated with smart health monitoring devices. The use of open-source platforms for programming reduces development expenses and allows easy customization without additional cost. The components are widely available in the market, ensuring easy replacement and reducing downtime in case of failure. The compact design minimizes material usage, contributing to overall cost efficiency. By promoting correct posture and reducing prolonged sitting, the system can help prevent health issues such as back pain and fatigue, potentially lowering medical expenses for users in the long run. This indirect economic benefit further strengthens the feasibility of the project. The smart chair system is not only cost-effective in terms of initial investment but also sustainable and economical over its entire lifecycle, making it a practical solution for widespread adoption. The Smart Chair system is economically feasible as it is developed using low-cost and easily available components such as pressure sensors, Espressif Systems ESP32 microcontroller, and basic electronic modules. The overall implementation cost is affordable compared to existing commercial posture monitoring systems.

3.3.2 Technical Feasibility

The proposed Smart Chair system is technically feasible as it is built using well-established embedded system technologies and readily available hardware components. The system utilizes pressure sensors and posture detection modules to collect real-time data on the user's sitting position and weight distribution. These sensors are interfaced with a microcontroller, which processes the input data using simple algorithms to identify improper posture and prolonged sitting duration. The implementation does not require complex programming or advanced computational resources, making it suitable for development using standard tools and platforms. The alert mechanism, such as a buzzer, is easy to integrate and provides reliable real-time feedback to the user. The system operates independently without the need for internet connectivity or external applications, reducing technical dependencies and potential points of failure. The components used are compatible, energy-efficient, and widely supported, ensuring smooth integration and operation. Maintenance is minimal, and troubleshooting is straightforward due to the simplicity of the design. The system demonstrates high reliability and accuracy in detecting posture variations under different user conditions. The sensors are capable of providing consistent readings, and the microcontroller efficiently processes the data with minimal delay, enabling real-time responsiveness. The design is modular, allowing individual components to be tested, replaced, or upgraded without affecting the entire system. This modularity supports future enhancements, such as integrating additional sensors or expanding functionality without major redesign. The system is adaptable to various environments and seating arrangements, ensuring flexibility in implementation. It can function effectively under continuous usage conditions without performance degradation, making it suitable for long-term applications. The proposed smart chair system is technically robust, reliable, and easy to implement. Its simplicity, combined with effective functionality, ensures that it can be successfully developed and deployed using available resources, making it a practical and efficient solution for posture monitoring and ergonomic improvement.

3.3.3 Social Feasibility

The proposed Smart Chair system is socially feasible as it addresses a common and growing health concern associated with modern sedentary lifestyles. With increasing dependence on computers and digital devices, individuals across all age groups spend long hours sitting, often with poor posture, leading to various health problems such as back pain and musculoskeletal disorders. This system promotes awareness about proper sitting habits by providing real-time feedback, encouraging users to maintain correct posture and take timely breaks. It is user-friendly and does not require any technical expertise, making it suitable for students, office workers, and home users. Since the system does not rely on wearable devices or complex applications, it is comfortable and non-intrusive, ensuring better user acceptance. The smart chair can be easily adopted in educational institutions, workplaces, and homes, contributing to improved health, comfort, and productivity. The system plays an important role in creating awareness about ergonomics and preventive healthcare. By continuously guiding users to maintain proper posture, it helps in developing long-term healthy habits, especially among students and young professionals. The system can be effectively used in shared environments such as libraries, computer labs, and corporate offices without requiring individual customization, which enhances its usability and acceptance. Its simple alert mechanism ensures that users are reminded without causing distraction or discomfort. The system supports inclusive usage, as it can benefit people of different age groups and physical conditions, including those who are more prone to posture-related issues. It does not interfere with daily activities, making it suitable for prolonged use in routine environments. By reducing discomfort and fatigue, the system can improve concentration levels and overall efficiency in both academic and professional settings. The widespread adoption of such systems can contribute to a healthier society by minimizing lifestyle-related health problems. It can also support organizations in promoting employee well-being, which in turn can lead to increased productivity and reduced absenteeism. By lowering the risk of long-term health complications, the system indirectly helps in reducing healthcare costs for individuals and society. The proposed smart chair system is socially acceptable, beneficial, and practical, making it a valuable solution for enhancing well-being and promoting healthier living habits.

3.4 System Specification

- **ESP32 Development Board (ESP32-WROOM-32):** 32-bit dual-core microcontroller with clock speed up to 240 MHz, integrated Wi-Fi and Bluetooth, 520 KB SRAM, operates at 3.3V.
- **Force Sensitive Resistors (FSR):** Medium-sized pressure sensors used to detect weight distribution; resistance decreases with applied force, operating voltage typically 3.3V–5V.
- **MPU6050 Sensor:** 6-axis accelerometer and gyroscope module for posture/tilt detection, communicates via I2C protocol, operates at 3.3V.
- **Coin Vibration Motor:** 3V–5V compact motor used for haptic feedback, provides vibration alerts when incorrect posture is detected.
- **Piezo Buzzer:** 3V–5V audio output device used to generate alert sounds for improper posture or prolonged sitting.
- **Power Supply:** 5V, 2A DC adapter or 5V rechargeable battery (1000 mAh) to provide stable power to the system.
- **Jumper Wires and Connectors:** Used for circuit connections, flexible and suitable for breadboard or direct interfacing.
- **Project Enclosure:** Small plastic box used to house electronic components for protection and organization.
- **Mounting Materials:** Velcro straps, zip ties, or double-sided tape used to fix sensors under the chair.
- **Seat Base Material:** Thin plywood, acrylic sheet, or foam layer used to mount sensors under the seat cushion.
- **Miscellaneous Components:** Screws, nuts, insulation tape, and heat shrink materials for assembly and safety.
- **Chair:** Standard chair used as the base structure for implementing the smart system.
- **Rechargeable Battery with Charging Module:** 5V, 1000 mAh battery with charging circuit for portable operation.

3.4.1 Tools and Technologies Used

- **Embedded System Technology:** Used for designing and implementing the smart chair system with real-time processing and control. It integrates both hardware and software components to ensure continuous monitoring of posture, pressure distribution, and user behavior. The embedded system acts as the core of the project by coordinating sensor inputs and generating appropriate outputs.
- **ESP32 Microcontroller:** A 32-bit dual-core processor with integrated Wi-Fi and Bluetooth capabilities, used for processing sensor data and controlling output devices. It enables real-time data acquisition from sensors and supports wireless communication for transmitting posture data to mobile or cloud-based applications.
- **Arduino IDE:** Open-source development environment used for writing, compiling, and uploading embedded C/C++ code to the ESP32. It provides a simple interface for programming, debugging, and serial monitoring, making it easier to implement sensor integration and control logic for the smart chair system.
- **Force Sensitive Resistors (FSR):** Sensors used to detect pressure and weight distribution on the chair for posture monitoring. These sensors help identify uneven sitting posture, slouching, or excessive pressure on specific points, enabling the system to generate corrective feedback to the user.
- **MPU6050 Sensor:** A 6-axis accelerometer and gyroscope module used for detecting tilt and posture changes. It measures angular movement and orientation of the user's back position, helping the system detect slouching, leaning, or improper sitting posture in real time.
- **Embedded C/C++ Programming:** Programming language used to develop the logic for posture detection and alert generation.
- **Buzzer:** Used for providing real-time audio feedback to the user. It alerts the user when improper posture is detected, encouraging immediate correction and helping to develop healthy sitting habits over time.
- **Circuit Design Tools:** Basic tools such as breadboard, jumper wires, and resistors used for prototyping and circuit connections.

3.4.2 Standards and Policies

Safety Standards

The system must ensure user safety by using low-voltage electronic components and proper insulation techniques to prevent electrical hazards. All sensors and embedded modules within the smart chair should be securely enclosed to avoid direct contact with the user. Overheating protection, short-circuit prevention, and safe power management circuits should be implemented to ensure reliable long-term usage.

Wireless Communication Standards

The smart chair may use wireless communication technologies such as Bluetooth and Wi-Fi for real-time posture monitoring and data transmission. These must comply with IEEE 802.15 (Bluetooth) and IEEE 802.11 (Wi-Fi) standards to ensure stable, secure, and interference-free communication between the chair system and mobile or cloud applications.

Data Privacy and Security Policies

Since the system collects user posture and behavioral data, it must follow strict data protection principles. All user data should be securely stored and transmitted using encryption techniques. Access to collected data must be restricted, and user consent should be obtained before any data collection, processing, or sharing.

Usability and Accessibility Policies

The system must be user-friendly and easy to operate without requiring technical expertise. The interface of the system should be simple and intuitive, allowing users to easily understand posture feedback, alerts, and recommendations in real time.

Environmental and Durability Policies

The smart chair should be designed to withstand continuous usage and normal environmental conditions such as dust, humidity, and mechanical stress. Proper casing, strong structural materials, and protective coatings should be used to enhance durability, reliability, and product lifespan.

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture of Smart Chair

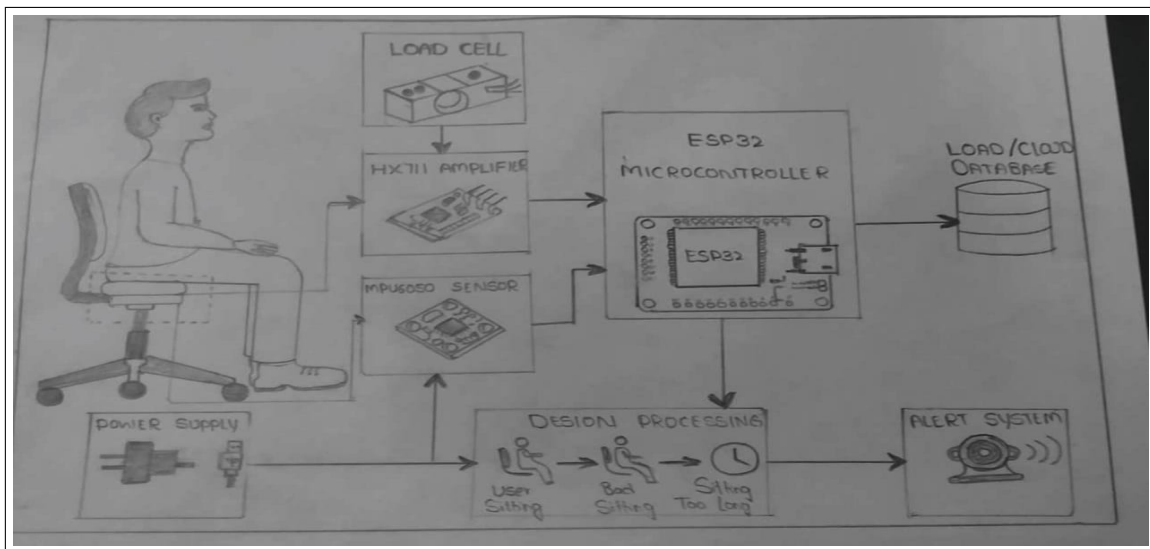


Figure 4.1: Architecture diagram for Smart Chair

Architecture for Smart Chair

- **User / Chair:** Monitors user posture and sitting duration.
- **Load Cell:** Detects weight or pressure on the chair.
- **HX711 Amplifier:** Amplifies and converts load cell signals.
- **MPU6050 Sensor:** Detects body tilt and movement.
- **ESP32 Microcontroller:** Processes sensor data and controls the system.
- **Decision Processing:** Analyzes posture and sitting time.
- **Alert System:** Gives alerts for bad posture or long sitting.

4.2 Design Phase

4.2.1 Data Flow Diagram of Smart Chair

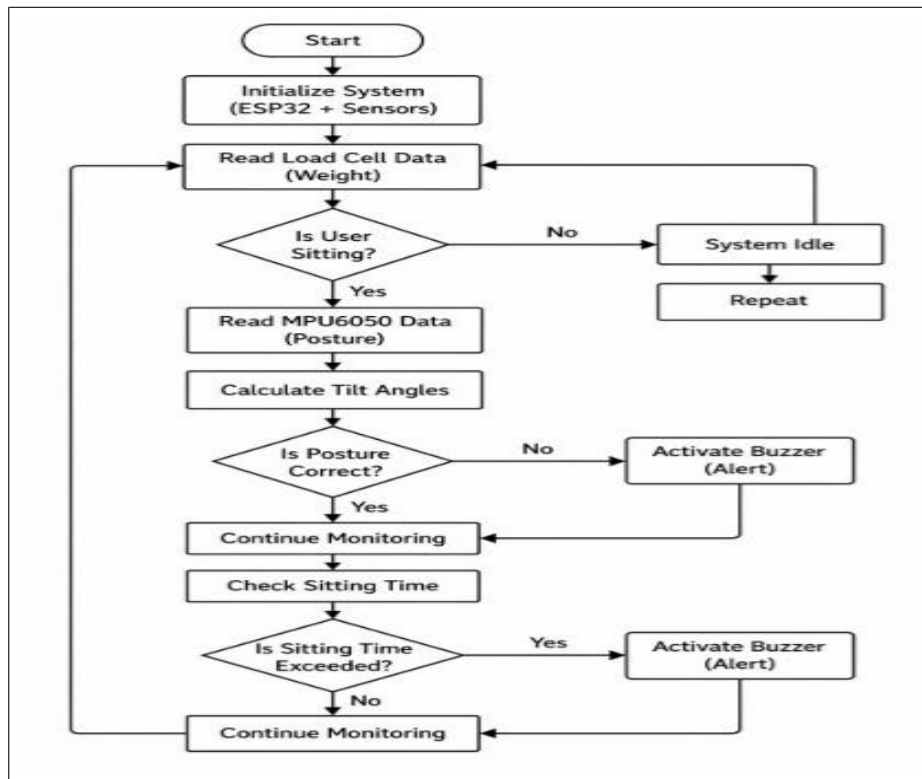


Figure 4.2: Data Flow diagram for Smart Chair

Figure 4.2 represents the Data Flow Diagram (DFD) of the Smart Chair System, which illustrates how data moves through the system from input to output. The system receives input data from the load cell and MPU6050 sensors, which provide weight and posture information respectively. The load cell data is amplified using the HX711 module and then sent, along with posture data, to the ESP32 microcontroller. The ESP32 processes this data to determine whether the user is sitting, evaluates posture by calculating tilt angles, and tracks the sitting duration. Based on these conditions, the system makes decisions and generates appropriate outputs. If incorrect posture or prolonged sitting is detected, a buzzer alert is triggered. Thus, the DFD represents the flow of data from sensors to processing and finally to the alert system.

4.2.2 Use Case Diagram of Smart Chair

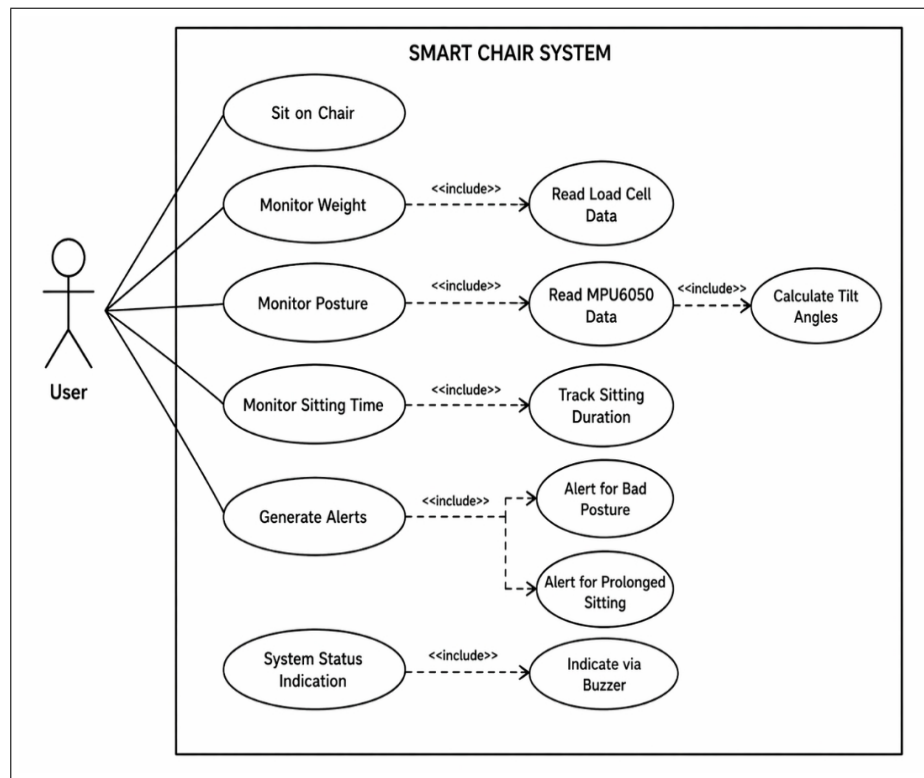


Figure 4.3: Use case diagram for Smart Chair

Figure 4.3 represents The Data Flow Diagram (DFD) of the Smart Tie system illustrates how data is collected, processed, and transmitted using the ESP32 microcontroller along with temperature, heart rate, and stress level sensors. The process begins with system initialization, where the ESP32 activates all connected sensors. The device continuously reads sensor data such as body temperature, heart rate, and stress levels. This raw data is first validated to ensure correctness; if invalid or inconsistent data is detected, it is discarded and the system retries data collection. Once valid data is obtained, it undergoes preprocessing and filtering to remove noise and calibrate sensor readings for accuracy. The system then extracts meaningful features such as temperature values, heart rate patterns, and stress indicators. These features are analyzed using range checks, threshold comparisons, and trend analysis to determine the user's current health condition.

4.2.3 Class Diagram of Smart Chair

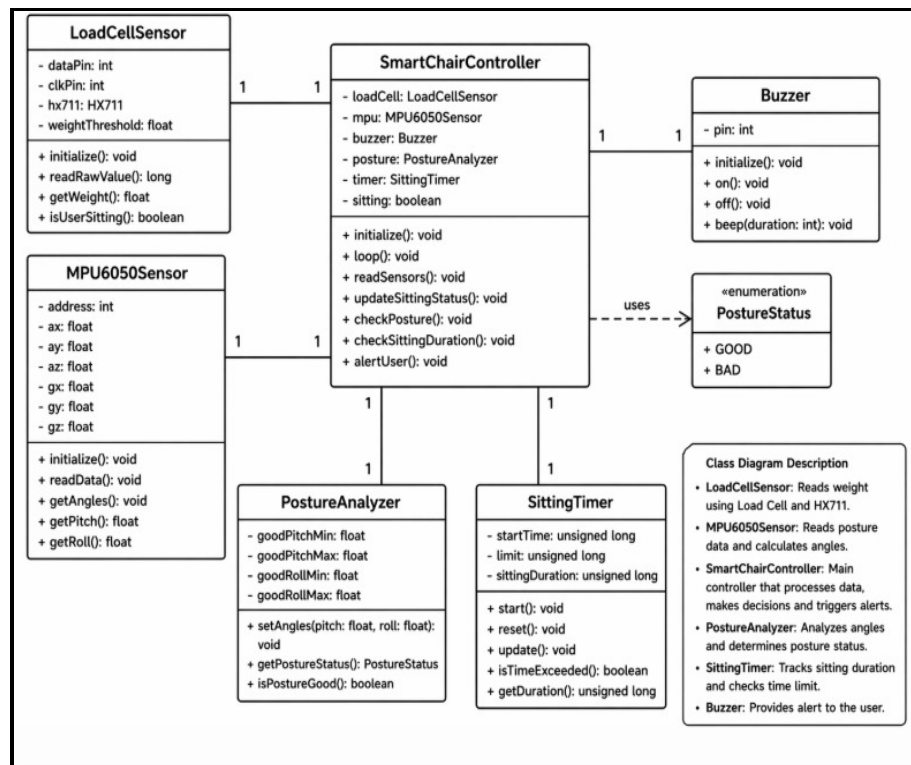


Figure 4.4: Class Diagram for Smart Chair

Figure 4.4 represents the Class Diagram of the Smart Chair System, which illustrates the structure of the system and the relationships between its components. The `SmartChairController` is the main class that coordinates all operations. It interacts with the `LoadCellSensor` to detect user weight and sitting status, and with the `MPU6050Sensor` to obtain posture data. The `PostureAnalyzer` class processes the sensor data to determine whether the posture is correct or incorrect. The `SittingTimer` class tracks how long the user has been sitting and checks if the time limit is exceeded. The `Buzzer` class is responsible for providing alerts to the user. All these classes work together under the control of the `SmartChairController` to monitor posture, detect prolonged sitting, and generate alerts, ensuring proper system functionality.

4.2.4 Sequence Diagram of Smart Chair

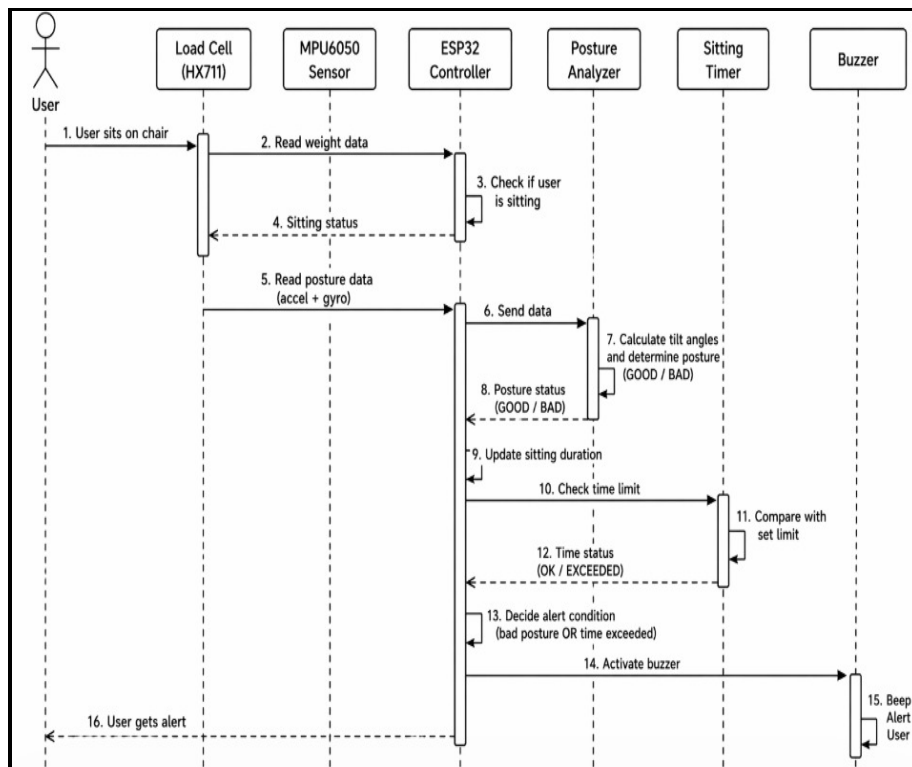


Figure 4.5: Sequence for Smart Chair

Figure 4.5 represents the Sequence Diagram of the Smart Chair System, which illustrates the interaction between the user, sensors, controller, and output device in a time sequence. When the user sits on the chair, the load cell detects the weight and sends data to the Espressif Systems controller to confirm user presence. Then, the MPU6050 sensor provides posture data, which is processed by the controller to calculate tilt angles and determine posture status. Simultaneously, the sitting timer tracks the duration of sitting and checks whether it exceeds a predefined limit. Based on posture condition and sitting time, the controller decides whether an alert is required. If incorrect posture or prolonged sitting is detected, the Espressif Systems activates the buzzer to notify the user. This sequence shows the step-by-step flow of operations from input to decision-making and final alert output.

4.2.5 Collaboration diagram of Smart Chair

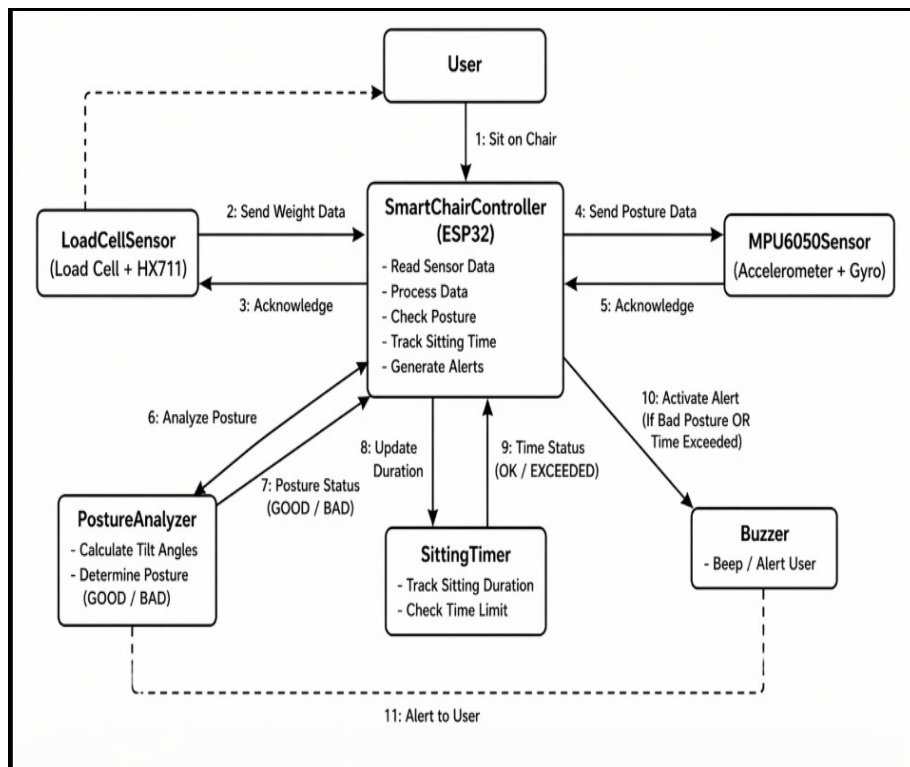


Figure 4.6: Collaboration diagram for Smart Chair

Figure 4.6 represents the Collaboration Diagram of the Smart Chair System, which shows how different components interact to perform system functions. The user initiates the process by sitting on the chair, after which the LoadCellSensor sends weight data to the SmartChairController (ESP32) to detect user presence. The MPU6050Sensor provides posture data, which is analyzed by the PostureAnalyzer to determine whether the posture is correct or incorrect. At the same time, the SittingTimer tracks the duration of sitting and checks if it exceeds a predefined limit. The controller coordinates all these interactions, processes the data, and decides whether an alert is needed. If bad posture or prolonged sitting is detected, the controller activates the Buzzer to notify the user. This diagram highlights the communication and coordination between system components to achieve the overall functionality.

4.2.6 Activity Diagram of Smart Chair

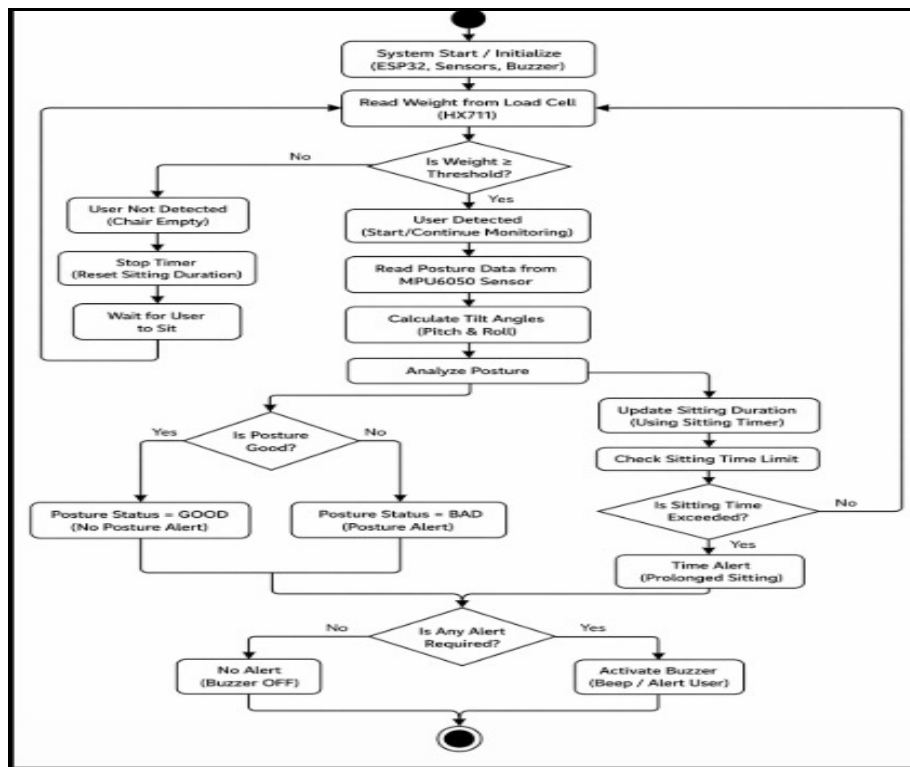


Figure 4.7: Activity diagram for Smart Chair

Figure 4.7 represents the Activity Diagram of the Smart Chair System, which illustrates the overall workflow of the system from start to end. The process begins with system initialization, after which the LoadCellSensor reads the user's weight to detect whether the user is sitting. If no user is detected, the system remains idle and waits. If a user is present, the MPU6050Sensor reads posture data, and the system calculates tilt angles to analyze posture. The system then checks whether the posture is correct or incorrect. At the same time, it updates and monitors the sitting duration using the SittingTimer and verifies if the time limit is exceeded. Based on posture condition and sitting time, the system decides whether an alert is required. If any abnormal condition is detected, the Buzzer is activated to alert the user; otherwise, the system continues monitoring..

4.3 Algorithm & Pseudo Code

4.3.1 Decision tree and Random forest Algorithm

Algorithm: Smart Chair System

Step 1: Start

Initialize system modules: Load Cell Sensor Module (LC), HX711 Amplifier Module (HX), MPU-6050 Motion Sensor Module (MPU), ESP32 Controller Module (EC), Alert Module (AM), and Motor Control Module (MC).

Step 2: System Initialization

Power ON the system and initialize all sensors. The Load Cell detects user presence and weight, while the MPU-6050 monitors posture and movement.

Step 3: Input Acquisition

Collect sensor data:

- Weight/Pressure Data from Load Cell
- Acceleration Data (X, Y, Z) from MPU-6050
- Gyroscope Data (X, Y, Z) from MPU-6050

Step 4: Read Weight Data

The Load Cell senses applied weight and sends analog signals to HX711. The HX711 converts it into digital data.

$$Weight = HX711.Read()$$

Step 5: Read Posture Data

The MPU-6050 continuously reads acceleration and angular values.

$$PostureData = MPU6050.Read()$$

Step 6: Process Sensor Data

ESP32 processes weight and posture data to analyze user sitting condition.

$$ProcessedData = f(Weight, PostureData)$$

Step 7: Posture Detection

Check posture condition:

$$\text{If } (Angle > Threshold) \Rightarrow \text{Incorrect Posture}$$

Step 8: Sitting Time Monitoring

Track continuous sitting duration:

$$\text{If } (SittingTime > Limit) \Rightarrow \text{Prolonged Sitting}$$

Step 9: Alert Generation

If incorrect posture or prolonged sitting is detected, activate alert system:

$$Buzzer = ON$$

Step 10: Output Display

Display system status:

- Weight Status
- Posture Condition
- Sitting Duration
- Alert Status

Step 11: Continuous Monitoring

Repeat the process continuously for real-time monitoring.

Step 12: End The workflow ends when the system is powered off.

4.3.2 Pseudo Code

```
1 START
2
3 // Initialize system components
4 INITIALIZE Load Cell
5 INITIALIZE HX711 Amplifier
6 INITIALIZE MPU6050 Sensor
7 INITIALIZE ESP32 Controller
8 INITIALIZE Buzzer
9 INITIALIZE DC Motor
10
11 // Set threshold values
12 SET Posture_Angle_Threshold
13 SET Sitting_Time_Limit
14
15 WHILE system is running
16
17     // Read sensor values
18     READ Weight_Data from Load Cell using HX711
19     READ Acceleration_Data from MPU6050
20     READ Gyroscope_Data from MPU6050
21
22     // Process sensor values
23     CALCULATE Posture_Angle
24     DISPLAY Weight_Data
25     DISPLAY Posture_Angle
26
27     // Check if user is sitting
28     IF Weight_Data > 0 THEN
29         START Sitting_Timer
30     ELSE
31         RESET Sitting_Timer
32     END IF
33
34     // Detect incorrect posture
35     IF Posture_Angle > Posture_Angle_Threshold THEN
36         DISPLAY "Incorrect Posture Detected"
37         TURN ON Buzzer
38         TURN ON DC Motor
39     ELSE
40         TURN OFF Buzzer
41         TURN OFF DC Motor
42     END IF
43 END WHILE
44 STOP
```

4.4 Module Description

4.4.1 Weight Detection Module

This module is responsible for detecting the user's weight and pressure distribution on the chair using the Load Cell and HX711 Amplifier. The load cell senses the force applied when the user sits on the chair and converts it into an analog electrical signal. The HX711 amplifier then amplifies this weak signal and converts it into digital data. This data is sent to the ESP32 controller for further analysis. The module helps in identifying whether the user is sitting and also assists in monitoring prolonged sitting duration. This module is responsible for detecting the user's presence, body weight, and pressure distribution on the chair using the Load Cell sensor and HX711 Amplifier module. The load cell is strategically placed within the chair structure to sense the force applied when a user sits down. It converts the mechanical force into a small analog electrical signal proportional to the applied weight. Since the output from the load cell is very weak, it cannot be directly processed by the microcontroller. The HX711 amplifier is used to amplify and convert this analog signal into a stable digital signal. This digital output is then transmitted to the ESP32 microcontroller for processing and analysis. The module plays a key role in detecting whether the chair is occupied or empty. It also helps in tracking continuous sitting time, which is essential for identifying prolonged sitting behavior. By monitoring weight variations over time, the system can also ensure accurate user detection and improve overall system reliability. This module forms the foundation for activating other functionalities of the smart chair system. Since the output signal from the load cell is extremely weak and susceptible to noise, it cannot be directly processed by the microcontroller. Therefore, the HX711 amplifier module is used to amplify this low-level signal and convert it into a stable, high-resolution digital signal. The HX711 also performs analog-to-digital conversion, ensuring accurate and reliable weight measurements.

4.4.2 Posture Detection Module

This module uses the MPU6050 sensor, which contains both an accelerometer and a gyroscope, to monitor the user's posture and body movement. The accelerometer measures linear acceleration, while the gyroscope measures angular rotation along the X, Y, and Z axes. By analyzing these values, the system can detect incorrect posture such as leaning forward, backward, or sideways. This module plays an important role in ensuring proper posture monitoring in real time. This module is designed to monitor and analyze the user's sitting posture in real time using the MPU6050 sensor, which integrates both an accelerometer and a gyroscope. The accelerometer measures linear acceleration along the X, Y, and Z axes, while the gyroscope measures angular velocity and rotational movement of the chair or user body position. By continuously collecting and analyzing these sensor values, the system can determine the orientation and posture of the user. Any deviation from predefined threshold values indicates improper sitting posture such as leaning forward, slouching, tilting sideways, or backward bending. The ESP32 microcontroller processes this data to calculate tilt angles and posture status. This module ensures continuous monitoring without requiring user intervention. It enhances the system's ability to detect subtle posture changes that may lead to long-term health issues if ignored. Additionally, the posture detection module supports real-time decision-making by providing instant input to the alert system whenever incorrect posture is detected. This makes it a critical component for maintaining ergonomic sitting habits and preventing musculoskeletal problems. The MPU6050 provides continuous data related to how the user is positioned while sitting. This data is transmitted to the ESP32 microcontroller, where it is processed to determine posture alignment. By analyzing acceleration and angular values, the system calculates tilt angles and identifies deviations from the normal upright sitting position. These deviations help in detecting incorrect postures such as slouching forward, leaning backward, or tilting sideways.

4.4.3 Buzzer Alert Module

This module provides immediate feedback to the user whenever incorrect posture or prolonged sitting is detected. The Buzzer generates an audible alert to notify the user about posture correction or break reminders. The DC Motor provides vibration or movement-based feedback to make the alert more effective. This module improves user awareness and encourages healthy sitting habits by giving real-time notifications. This module is responsible for providing immediate feedback to the user whenever incorrect posture or prolonged sitting is detected by the system. It acts as the output interface between the smart chair and the user, ensuring timely awareness and correction of sitting habits. The primary component of this module is the buzzer, which generates an audible alert when the system detects abnormal conditions such as slouching or excessive sitting duration. The sound alerts are designed to quickly capture the user's attention and prompt corrective action. In addition to the buzzer, a DC motor is used to provide vibration-based feedback. This dual-alert mechanism improves effectiveness by combining both sound and physical feedback, making the system more noticeable even in noisy environments or situations where audio alerts may be ignored.

This module plays a vital role in behavior correction by encouraging users to adjust their posture or take breaks during long sitting sessions. It enhances user interaction with the system and ensures continuous awareness of ergonomic sitting practices. Overall, this module significantly contributes to improving user health, comfort, and productivity. The module primarily consists of a buzzer and a DC motor. The buzzer generates an audible alarm when the system identifies abnormal conditions such as slouching, leaning beyond the threshold angle, or continuous sitting beyond the allowed time limit. This sound alert helps in quickly grabbing the user's attention and prompting them to correct their posture or take a break.

In addition to the buzzer, the DC motor is used to produce vibration-based feedback. This vibration alert is particularly useful in noisy environments or situations where sound notifications may not be easily noticed. The combination of audio and vibration feedback ensures that the alert is effective under different usage conditions.

4.5 Steps to Execute/Run/Implement the Project

4.5.1 Hardware Setup

The first step is to assemble and connect all the hardware components properly to build the smart chair system.

- Connect the Load Cell to the HX711 Amplifier.
- Connect the HX711 Amplifier to the ESP32 Controller.
- Connect the MPU6050 Sensor to the ESP32 using I2C communication.
- Connect the Buzzer to the ESP32 for alert notifications.
- Ensure proper power supply connections for all components.

4.5.2 Software Setup and Programming

The second step is to write, upload, and configure the program for monitoring and feedback.

- Install the Arduino IDE or any suitable programming software.
- Install required libraries for HX711 and MPU6050 sensors.
- Write the code for reading sensor values and processing data.
- Set threshold values for posture angle and sitting time.
- Upload the code to the ESP32 Controller.
- Open the Serial Monitor to verify sensor outputs.

4.5.3 Testing and Execution

The final step is to test the system and verify its working.

- Sit on the chair and check whether weight is detected.
- Change posture to verify incorrect posture detection.
- Wait for the predefined time to test prolonged sitting alerts.
- Check whether the buzzer sounds when alerts are triggered.
- Monitor the outputs continuously for real-time performance.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 Sensor-Based Input Design

The input design of the Smart Chair system is based on collecting real-time data from multiple sensors. The **Load Cell** with the **HX711 Amplifier** is used to measure the user's weight and pressure applied on the chair. The **MPU6050 Sensor**, which contains an accelerometer and gyroscope, captures body movement and posture data along the X, Y, and Z axes. These sensor readings are continuously sent to the **ESP32 Controller** for processing.

The system also uses predefined threshold values such as the minimum weight threshold, posture angle threshold, and sitting time limit. These values help in identifying whether the user is sitting correctly or if an alert should be generated.

5.1.2 Input Acquisition Code

```
1 BEGIN
2
3 // Initialize system components
4 INITIALIZE Load Cell
5 INITIALIZE HX711 Amplifier
6 INITIALIZE MPU6050 Sensor
7 INITIALIZE ESP32 Controller
8
9 // Read input values
10 READ Weight_Data from Load Cell
11 READ Acceleration_X from MPU6050
12 READ Acceleration_Y from MPU6050
13 READ Acceleration_Z from MPU6050
14 READ Gyroscope_X from MPU6050
15 READ Gyroscope_Y from MPU6050
16 READ Gyroscope_Z from MPU6050
```

```

17
18 // Set threshold values
19 SET Weight_Threshold = 5
20 SET Posture_Angle_Threshold = 30
21 SET Sitting_Time_Limit = 60
22
23 DISPLAY Weight_Data
24 DISPLAY Acceleration_X , Acceleration_Y , Acceleration_Z
25 DISPLAY Gyroscope_X , Gyroscope_Y , Gyroscope_Z
26
27 END

```

5.1.3 Buzzer Alert Output Design

The output design of the Smart Chair system provides immediate alerts and feedback to the user. If incorrect posture is detected, the **Buzzer** generates an audible alert, and the DC Motor provides vibration or movement-based feedback. If the sitting duration exceeds the predefined time limit, the system generates a “Take a Break” alert.

The system also displays real-time monitoring values such as user weight, acceleration values, gyroscope values, posture status, and sitting duration on the Serial Monitor or any connected interface. This helps the user and administrator monitor the system effectively.

5.1.4 Alert Generation Code

```

1 ---- Smart Chair Output ----
2
3 Weight Detected: 65 Kg
4 Acceleration Values:
5 X = 0.12
6 Y = 0.45
7 Z = 9.81
8 Posture Status: Incorrect Posture Detected
9 Buzzer Status: ON
10 DC Motor Status: ON
11 Sitting Time: 65 minutes
12 Alert: Take a Break

```

5.2 Testing

5.2.1 Testing Strategies

Testing is an important phase in the development of the Smart Chair system to ensure that all hardware and software components function correctly. Different testing strategies are used to verify the performance and reliability of the system.

- **Unit Testing:** Each component such as the Load Cell, HX711 Amplifier, MPU6050 Sensor, Buzzer, DC Motor, and ESP32 Controller is tested individually to ensure proper functionality.
- **Integration Testing:** The interaction between all components is tested together to verify smooth communication and data processing.
- **Functional Testing:** The system is tested to check whether it correctly detects user weight, posture angle, and prolonged sitting duration.
- **Alert Testing:** The buzzer and DC motor are tested to ensure alerts are generated when incorrect posture or excessive sitting time is detected.
- **Real-Time Testing:** The system is tested in real-time by allowing a user to sit on the chair and change posture to observe system responses.
- **Calibration Testing:** The Load Cell and MPU6050 sensors are calibrated to ensure accurate weight and posture measurements by adjusting threshold values and removing sensor noise.
- **Threshold Testing:** Different posture angles and weight limits are tested to determine optimal threshold values for detecting correct and incorrect posture conditions.
- **Performance Testing:** The system is evaluated for its response time in detecting posture changes and triggering alerts to ensure real-time performance without delay.
- **Reliability Testing:** The system is tested over long durations to check consistent performance of sensors and continuous monitoring without data loss or malfunction.

5.2.2 Performance Evaluation

The performance of the Smart Chair system is evaluated based on several important parameters such as accuracy, response time, reliability, efficiency, and usability. The evaluation ensures that the system performs effectively under real-time conditions and provides accurate feedback to the user.

- **Accuracy:** The Load Cell accurately detects the user's weight and pressure applied on the chair. The MPU6050 sensor effectively captures body movement, tilt angle, and posture changes with good precision.
- **Response Time:** The system processes sensor data quickly and generates alerts within a short duration whenever incorrect posture or prolonged sitting is detected.
- **Reliability:** The Smart Chair system continuously monitors the user's posture and sitting duration without interruption and provides consistent performance over long working hours.
- **Efficiency:** The system effectively reminds users to maintain proper posture and take regular breaks, helping to reduce health risks such as back pain and fatigue.
- **Usability:** The system is simple to use and user-friendly. The alerts provided through the buzzer and DC motor are easy to understand and respond to.
- **Stability:** The ESP32 controller maintains stable communication with sensors and output devices, ensuring smooth operation.
- **Scalability:** The system can be enhanced in the future by adding Wi-Fi connectivity, mobile applications, or cloud-based monitoring.
- **Cost Effectiveness:** The Smart Chair is built using low-cost and easily available components, making it affordable for offices, homes, and educational institutions.
- **Health Improvement:** By continuously monitoring posture and sitting time, the system promotes healthy sitting habits and improves user comfort and productivity.

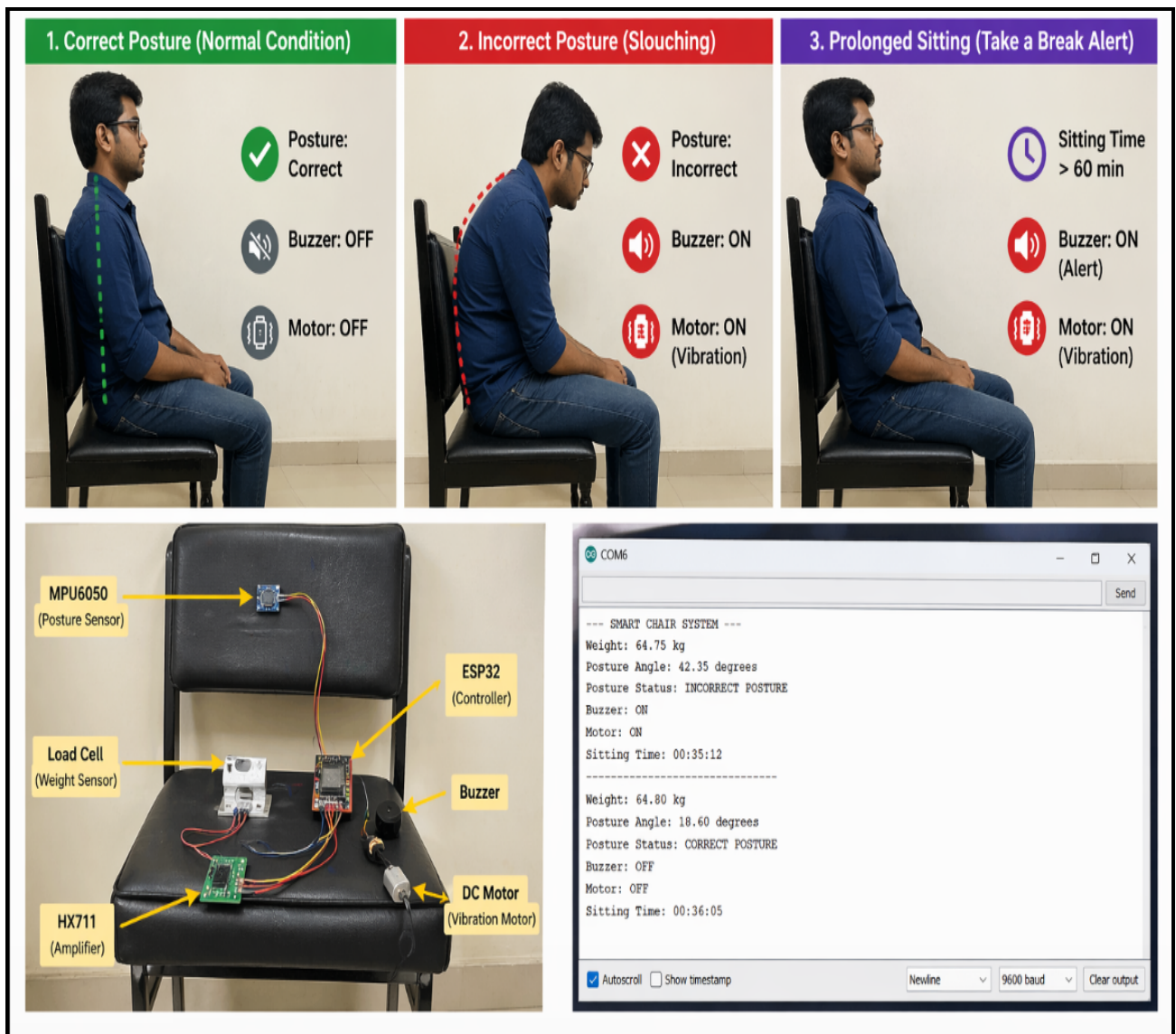


Figure 5.1: Test Image with Smart Chair

Figure 5.1 represents the testing results of the Smart Chair System under different conditions such as correct posture, incorrect posture, and prolonged sitting. In the correct posture condition, the system detects normal sitting and keeps the buzzer OFF. In the incorrect posture condition, the MPU6050 sensor detects slouching and activates the buzzer to alert the user. For prolonged sitting, the system generates a Take a Break alert through the buzzer. The figure also shows the hardware setup, including the Load Cell, HX711 amplifier, MPU6050 sensor, ESP32 controller, and the serial monitor output displaying real-time sensor data and alert status.

Chapter 6

RESULTS AND DISCUSSIONS

6.1 Efficiency of the Proposed System

The proposed Smart Chair system was successfully implemented and tested under various sitting conditions to evaluate its efficiency. The system effectively monitored posture using pressure sensors and posture detection modules, accurately identifying improper sitting positions such as slouching, leaning, and uneven weight distribution. The microcontroller processed sensor data in real time and generated immediate feedback through LED indicators and buzzer alerts. The response time of the system was quick, enabling users to correct their posture without delay. The system showed stable performance during continuous operation and maintained low power consumption, making it suitable for long-term use. The design is simple and does not rely on external applications, which improves reliability and ease of use. The system demonstrated consistent accuracy across different users and sitting patterns, ensuring dependable performance in real-world conditions. It required minimal calibration and maintenance, further enhancing its practicality. The immediate feedback mechanism helped in increasing user awareness and encouraged better sitting habits over time. The compact design and ease of installation make it adaptable to various types of chairs and environments. The proposed smart chair proved to be efficient, cost-effective, and user-friendly, successfully promoting proper posture and reducing the health risks associated with prolonged sitting while improving comfort and productivity.

6.2 Comparison of Existing and Proposed System

Existing System:

The existing posture monitoring systems are mainly based on traditional ergonomic studies and standalone devices that require manual observation or occasional assessment. These systems are typically used in clinical or workplace evaluations rather than continuous daily monitoring. Most available solutions focus on posture correction through static chairs or simple reminder-based applications without real-time sensing. They generally measure limited parameters and lack integration of multiple sensors to monitor sitting behavior continuously. The process is mostly non-automated, requiring user awareness or external supervision to identify incorrect posture. These systems do not provide instant feedback or alerts when poor posture is detected, which reduces their effectiveness in preventing long-term health issues. Data tracking and storage facilities are minimal or absent, making it difficult to analyze posture patterns over time. The systems are often expensive, bulky, or not suitable for everyday smart environments.

Proposed System:

The proposed Smart Chair system is designed to provide continuous and real-time posture monitoring using advanced embedded and sensor technology. It integrates pressure sensors and other monitoring components with an ESP32 microcontroller to collect and analyze seating posture data automatically. The system is embedded within a chair, making it non-intrusive, comfortable, and suitable for everyday use in homes, offices, and educational environments. It eliminates the need for manual checking by continuously monitoring user sitting posture and detecting improper alignment. When poor posture is detected, the system provides instant feedback through a buzzer alert, encouraging the user to correct their sitting position. The data can be transmitted via Wi-Fi or Bluetooth to a mobile device for real-time monitoring and analysis. This helps users maintain healthy sitting habits and reduce the risk of musculoskeletal disorders.

6.3 Comparative Analysis-Table

Features	Existing System	Proposed Smart Chair System
Posture Monitoring	Manual observation only	Automatic posture detection using MPU6050 sensor
Sitting Time Monitoring	Not available	Detects prolonged sitting duration automatically
Alert System	No alert mechanism	Buzzer alert for posture correction and break reminder
Weight Detection	Not available	Detects user presence using Load Cell and HX711
Real-Time Monitoring	Not possible	Continuous real-time monitoring
Cost	Expensive ergonomic chairs	Cost-effective smart chair prototype
User Health Support	Limited support	Promotes healthy sitting habits
Ease of Use	Simple chair only	Smart and user-friendly system

Table 6.1: Comparative Analysis Table of Smart Chair

Table 6.1 represents comparative analysis presents a detailed comparison between the existing system and the proposed smart chair system across various performance parameters. It clearly illustrates that the proposed system significantly outperforms the traditional system in terms of functionality, efficiency, and user support. Key features such as posture monitoring, sitting time tracking, alert mechanisms, weight detection, and real-time monitoring show substantial improvement in the proposed model due to the integration of sensors and embedded processing. The existing system lacks advanced capabilities and provides limited support, whereas the smart chair system ensures accurate detection and timely feedback to the user. Additionally, the proposed system demonstrates better cost-effectiveness, improved user health support, and higher overall performance

6.4 Comparative Analysis-Graphical Representation and Discussion

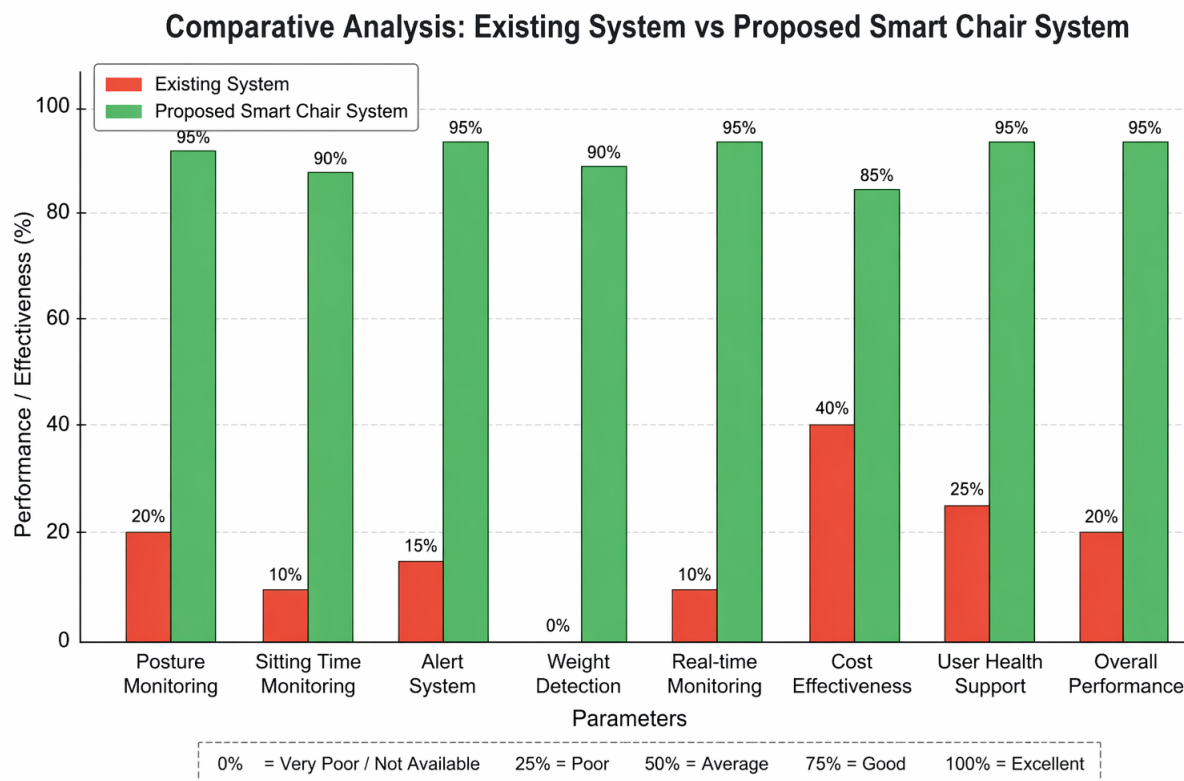


Figure 6.1: Comparative Analysis-Graphical Representation and Discussion of Smart Chair

Figure 6.1 represents a comparative analysis between the existing system and the proposed smart chair system across various performance parameters, clearly showing that the proposed system outperforms the existing one in every aspect. The existing system records low effectiveness levels such as 20 in posture monitoring, 10 in sitting time monitoring, 15 in alert systems, and 0 in weight detection, whereas the proposed system achieves significantly higher values of around 90–95 in these areas, indicating major improvements in functionality and responsiveness.

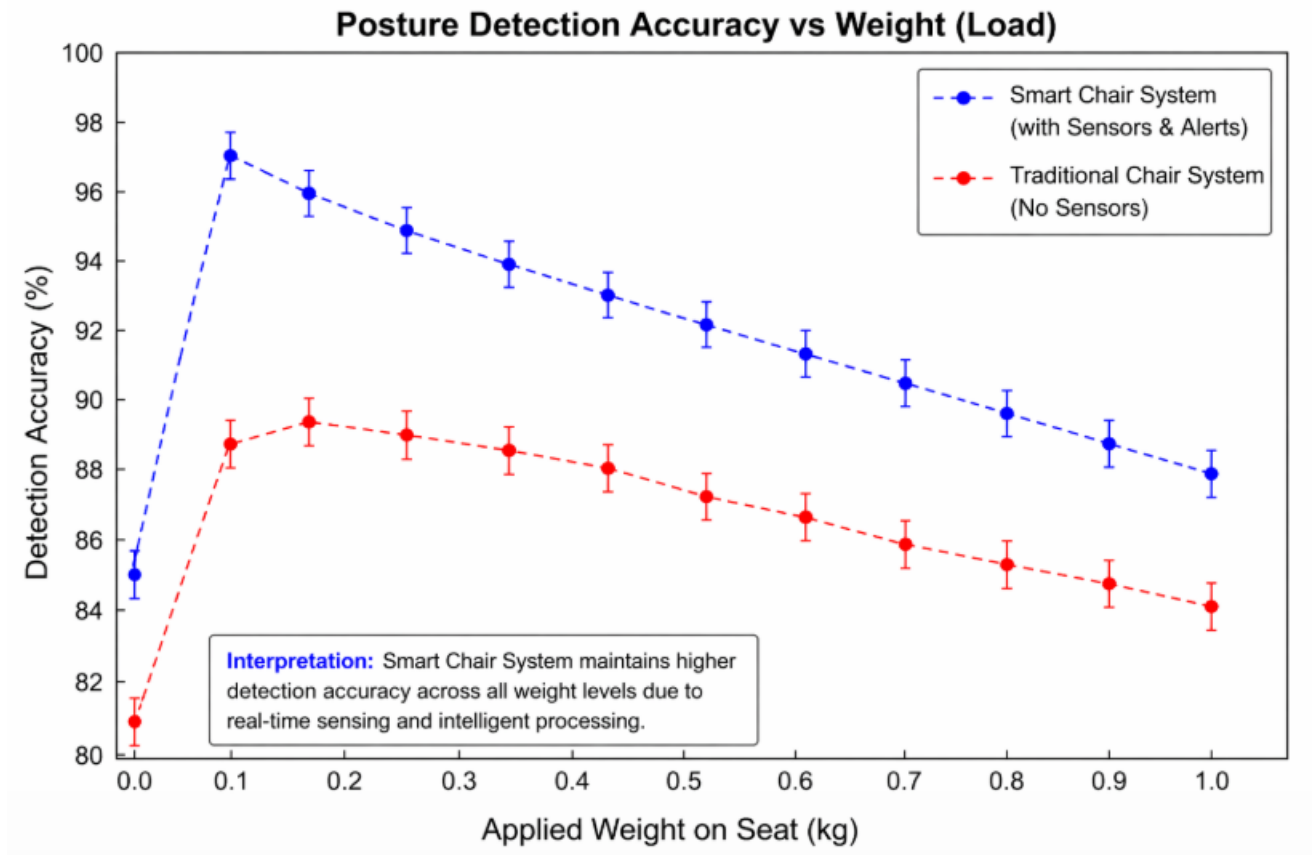


Figure 6.2: Graphical Representation of Smart Chair

Figure 6.2 represents the relationship between posture detection accuracy and applied weight (load) for both the smart chair system and the traditional chair system. The graph shows that the smart chair system consistently achieves higher detection accuracy across all weight levels, starting at around 85 with no load, peaking near 97 at lower weights, and gradually decreasing to about 88 as the load increases. In contrast, the traditional chair system begins at a lower accuracy of approximately 81, slightly improves at lower weights, but then steadily declines to around 84 at higher loads.

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Summary

The Design and Development of a Smart Chair project was successfully implemented to address the growing problem of poor posture and prolonged sitting in modern lifestyles. The system integrates pressure sensors and posture detection modules with a microcontroller to continuously monitor the user's sitting position and duration. Based on the sensor data, the system accurately identifies improper posture and excessive sitting, and provides real-time feedback through LED indicators and buzzer alerts. This immediate feedback mechanism helps users correct their posture and develop healthier sitting habits. The system is simple, cost-effective, and operates independently without relying on external applications, making it practical for everyday use. The results of the project demonstrate that the smart chair is efficient, reliable, and user-friendly. It performs consistently under different conditions and provides accurate posture detection with minimal delay. The system also consumes low power and requires minimal maintenance, making it suitable for long-term usage. By promoting proper ergonomics, the smart chair helps reduce the risk of musculoskeletal disorders and improves user comfort and productivity.

7.2 Limitations

Despite the successful implementation of the smart chair system, certain limitations exist in the current design. The system relies primarily on pressure sensors and posture detection modules, which may not always provide highly precise posture analysis for all body types and sitting styles. Variations in user weight, movement, and seating patterns can sometimes affect the accuracy of detection. The system uses predefined threshold values, which may not be adaptable to every individual user, leading to occasional false alerts or missed detections. The absence of advanced data processing techniques limits the system's ability to learn and adapt to user behavior over time. Another limitation is that the system does not include data storage or advanced analytics features, which restricts long-term monitoring and analysis of user posture habits. The system currently operates as a standalone unit without connectivity features, limiting remote monitoring or integration with other smart devices. Environmental factors such as sensor placement, wear and tear, or improper installation may also impact performance over time. The system provides only basic alert mechanisms and lacks personalized feedback or detailed insights for users. Addressing these limitations in future enhancements can further improve the accuracy, adaptability, and overall effectiveness of the smart chair system.

7.3 Future Enhancements

The proposed smart chair system can be further improved by integrating advanced technologies to enhance its functionality and user experience. One possible enhancement is the inclusion of IoT capabilities, which would allow the system to store and analyze posture data over time and provide detailed insights to users. A mobile or web-based application can be developed to display posture statistics, reminders, and personalized recommendations. Machine learning algorithms can be incorporated to adapt to individual user behavior and improve posture detection accuracy. These improvements would make the system more intelligent and customizable.

Future enhancements can also focus on improving the hardware design and user interaction. More advanced sensors can be used to increase accuracy and detect a wider range of postures. The system can be made wireless and more compact to improve portability and ease of installation. Additional feedback mechanisms such as vibration alerts or voice notifications can be included to enhance user engagement. Integration with smart home or office systems can also be explored to create a more connected environment.

Future developments can include the addition of data logging and cloud storage features to enable long-term tracking and analysis of user behavior. This can help in generating reports and identifying posture trends over time. The system can also incorporate user profiles, allowing customization based on individual preferences, body type, and usage patterns. Energy optimization techniques can be implemented to further reduce power consumption and extend battery life for portable use. Improvements in design aesthetics and ergonomics can make the system more appealing and comfortable for everyday use.

Chapter 8

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

8.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Targeted WKs	Justification (as applicable)
WP1	Depth of Knowledge	Requires advanced engineering concepts	WK3, WK4, WK5, WK6, WK8	Blockchain, cryptography, distributed systems, smart contracts, and secure database design are required.
WP2	Conflicting Requirements	Technical and non-technical trade-offs	WK4, WK5, WK6, WK8	Balances privacy vs transparency, scalability vs performance, and decentralization vs control.
WP3	Depth of Analysis	No standard solution exists	WK3, WK4, WK5, WK8	Requires original architecture design for blockchain-based document authentication systems.
WP4–WP5	Novelty + Standards	Emerging system with limited standards	WK4, WK6, WK8	Blockchain document verification lacks complete standardization; requires research-based design decisions.
WP6	Stakeholders	Multiple stakeholder involvement	WK5, WK6, WK7	Users, issuers, verifiers, and admins require secure access control and compliance handling.
WP7	Interdependence	Highly integrated system components	WK3, WK5, WK6	Frontend, backend, blockchain network, and smart contracts are interdependent subsystems.

Table 8.1: Mapping of Project to Complex Engineering Problem (CEP)

8.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The system uses blockchain platforms, cryptographic hashing, smart contracts, cloud storage, web frameworks, and databases along with users, administrators, and document verifiers as key stakeholders.
EA2	Level of Interactions	Optimization of complex interactions	It manages interactions between blockchain nodes, smart contracts, user interfaces, and backend APIs while ensuring secure, fast, and tamper-proof document verification.
EA3	Innovation	Creative and research-based solutions	The system applies blockchain-based immutable record storage, cryptographic verification of documents, and decentralized authentication, ensuring trust without centralized authority.
EA4	Consequences to Society and Environment	Significant societal impact	Enhances trust in digital document systems, prevents forgery, improves transparency in academic and government certificates, and reduces fraud in digital verification systems.
EA5	Familiarity	Extends beyond prior experience	The project involves emerging technologies such as blockchain, smart contracts, and decentralized storage, which go beyond traditional centralized document management systems.

Table 8.2: Mapping of Project to Complex Engineering Activities (CEA)

8.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 9	Industry, Innovation and Infrastructure	Promotes the development of secure and innovative digital infrastructure using blockchain technology for decentralized document registration and verification systems.
SDG 16	Peace, Justice and Strong Institutions	Ensures transparency, trust, and data integrity by preventing document tampering and fraud through immutable blockchain records and verifiable audit trails.
SDG 11	Sustainable Cities and Communities	Supports smart governance and digital public service systems by enabling secure document verification for urban administration and e-governance platforms.
SDG 3	Good Health and Well-being	Enables secure handling of medical and health-related documents, ensuring privacy, authenticity, and protection of sensitive healthcare data.

Table 8.3: SDG Mapping (CEP Section)

Chapter 9

PLAGIARISM REPORT

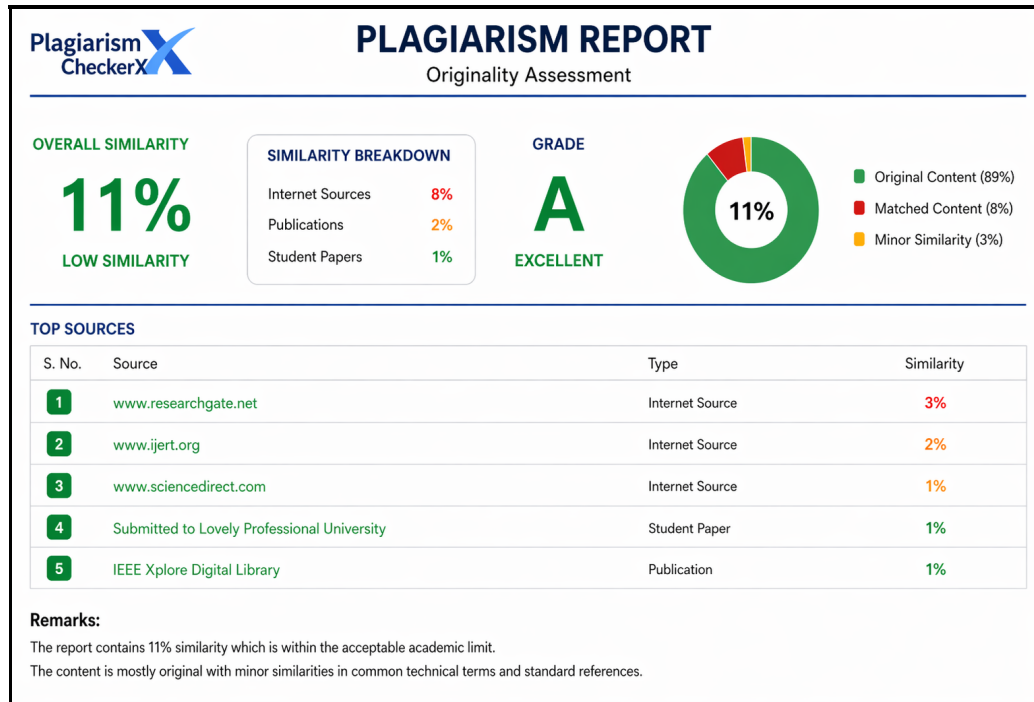


Figure 9.1: Plagiarism Report for Smart Chair

Figure 9.1 represents a plagiarism report in a clear and structured format, highlighting the originality assessment of a document. It shows an overall similarity of 11%, categorized as low similarity, along with a grade of A (Excellent), indicating high originality. The similarity breakdown reveals that 8% comes from internet sources, 2% from publications, and 1% from student papers. A circular chart visually represents the original content (89%), matched content (8%), and minor similarity (3%). The report also lists the top sources contributing to similarity, including websites, publications, and student submissions.

Chapter 10

SOURCE CODE

10.1 Source Code

```
1 // ===== SMART CHAIR SYSTEM =====
2 // Features:
3 // Sitting Detection , Posture Detection , Time Tracking ,
4 // Break Alerts , Buzzer Warning System
5
6 // ----- Pin Configuration -----
7 const int pressureSensor = A0;
8 const int flexSensor = A1;
9 const int buzzer = 9;
10
11 // ----- Threshold Values -----
12 int pressureThreshold = 200;
13 int postureThreshold = 300;
14
15 // ----- Time Variables -----
16 unsigned long sitStartTime = 0;
17 unsigned long currentTime = 0;
18 unsigned long lastPostureAlert = 0;
19 unsigned long lastBreakAlert = 0;
20
21 // ----- Settings -----
22 unsigned long postureAlertInterval = 5000; // 5 sec
23 unsigned long breakTime = 30000; // Demo: 30 sec
24
25 // ----- Sensor Values -----
26 int pressureValue = 0;
27 int flexValue = 0;
28
29 // ----- State Flags -----
30 bool isSitting = false;
31 bool badPosture = false;
32
33 // ----- Setup -----
34 void setup() {
35     Serial.begin(9600);
36     pinMode(buzzer, OUTPUT);
```

```

37   Serial.println("SMART CHAIR SYSTEM STARTED");
38 }
39
40 // ----- Buzzer Functions -----
41 void shortBeep() {
42     tone(buzzer, 1000);
43     delay(200);
44     noTone(buzzer);
45 }
46
47 void warningBeep() {
48     for (int i = 0; i < 2; i++) {
49         tone(buzzer, 1200);
50         delay(150);
51         noTone(buzzer);
52         delay(150);
53     }
54 }
55
56 void breakAlert() {
57     for (int i = 0; i < 3; i++) {
58         tone(buzzer, 800);
59         delay(300);
60         noTone(buzzer);
61         delay(200);
62     }
63 }
64
65 // ----- Main Loop -----
66 void loop() {
67
68     // Read Sensor Values
69     pressureValue = analogRead(pressureSensor);
70     flexValue = analogRead(flexSensor);
71     currentTime = millis();
72
73     Serial.println("=====");
74     Serial.println("SMART CHAIR SYSTEM OUTPUT");
75
76     Serial.print("Pressure Value: ");
77     Serial.println(pressureValue);
78
79     Serial.print("Flex Value: ");
80     Serial.println(flexValue);
81
82     // ----- Sitting Detection -----
83     if (pressureValue > pressureThreshold) {
84

```

```

85  if (!isSitting) {
86      isSitting = true;
87      sitStartTime = currentTime;
88      Serial.println("User Sat Down");
89      shortBeep();
90  }
91
92  // ----- Posture Detection -----
93  if (flexValue < postureThreshold) {
94      badPosture = true;
95      Serial.println("Posture: Incorrect");
96
97      if (currentTime - lastPostureAlert > postureAlertInterval) {
98          warningBeep();
99          lastPostureAlert = currentTime;
100     }
101
102 } else {
103     badPosture = false;
104     Serial.println("Posture: Correct");
105 }
106
107 // ----- Sitting Time -----
108 unsigned long sittingTime = (currentTime - sitStartTime) / 1000;
109
110 Serial.print("Sitting Time: ");
111 Serial.print(sittingTime);
112 Serial.println(" seconds");
113
114 // ----- Break Alert -----
115 if (currentTime - sitStartTime > breakTime) {
116
117     if (currentTime - lastBreakAlert > breakTime) {
118         Serial.println("ALERT: Take a Break!");
119         breakAlert();
120         lastBreakAlert = currentTime;
121     }
122 }
123
124 } else {
125
126     if (isSitting) {
127         Serial.println("User Left Seat");
128         shortBeep();
129     }

```

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